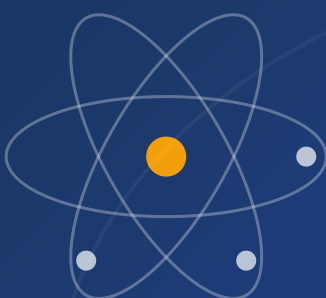


B.TECH FIRST YEAR · ENGINEERING CHEMISTRY

Structure & Bonding Models

Unit I – Complete Study Module

Quantum Mechanics · Schrödinger Equation · Wave Functions
Particle in a Box · Molecular Orbital Theory · Bond Order
 π -Molecular Orbitals · Butadiene · Benzene



Comprehensive · Step-by-Step · Exam-Oriented · 40+ Solved Problems

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Fully structured module covering all syllabus topics with diagrams, problems & Q&A

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How to Use This Module

Read each section in order. Every concept starts from basics with step-by-step explanation. Use the formula boxes for quick recall, diagrams for visual understanding, and solved problems for exam practice.

Introduction to Atomic Structure & Failure of Classical Mechanics

Why classical physics failed and why we needed quantum theory

1.1 What is Atomic Structure?

Every substance around us — water, iron, air — is made of **atoms**. Understanding how electrons are arranged around the nucleus, and how atoms bond together to form molecules, is the foundation of chemistry. This understanding comes from **quantum mechanics**.

1 Basic Idea – What is an Atom?

An atom has a positively charged **nucleus** (protons + neutrons) surrounded by negatively charged **electrons** that move in specific regions called orbitals. The arrangement of electrons determines chemical properties.

2 Classical Picture – Rutherford's Model (1911)

Rutherford proposed electrons orbit the nucleus like planets orbit the sun. This model **failed** because classical physics says a charged particle in circular motion radiates energy → the electron should spiral into the nucleus within 10^{-8} seconds. Atoms are stable → classical physics is wrong!

1.2 Failures of Classical Mechanics

Phenomenon	Classical Prediction	Experimental Observation	Conclusion
Blackbody Radiation	Energy emitted continuously (ultraviolet catastrophe)	Energy emitted in discrete packets	Energy is quantized (Planck, 1900)
Photoelectric Effect	Light intensity determines electron ejection	Light frequency determines ejection; intensity controls number	Light has particle nature (Einstein, 1905)
Hydrogen Spectrum	Continuous spectrum expected	Discrete line spectrum observed	Electrons occupy fixed energy levels

Compton Effect	No momentum expected for waves	X-rays scatter with changed wavelength	Photons carry momentum
Atomic Stability	Electrons should spiral into nucleus	Atoms are perfectly stable	Classical mechanics is invalid at atomic scale

1.3 Planck's Quantum Hypothesis (1900)



The Key Insight

Planck proposed that energy is not continuous — it comes in small indivisible packets called **quanta (photons)**. The energy of each quantum depends only on the frequency of radiation.

FORMULA

$$E = h\nu$$

where:

E = energy of one quantum (joules, J)

h = Planck's constant = 6.626×10^{-34} J·s

ν (nu) = frequency of radiation (Hz = s⁻¹)

1.4 de Broglie's Wave-Particle Duality (1924)

Louis de Broglie proposed that **all matter has wave nature**. Just as light behaves as both wave and particle, electrons (and all particles) also exhibit dual nature. This was experimentally confirmed by Davisson-Germer electron diffraction experiment.

FORMULA

$$\lambda = h / mv = h / p$$

where:

λ (lambda) = de Broglie wavelength of the particle

h = Planck's constant = 6.626×10^{-34} J·s

m = mass of particle (kg)

v = velocity of particle (m/s)

$p = mv$ = momentum of particle (kg·m/s)



Important Insight

For macroscopic objects (like a cricket ball), λ is negligibly small ($\sim 10^{-34}$ m), so wave nature is undetectable. For electrons (mass $\sim 10^{-31}$ kg), λ is comparable to atomic dimensions → wave nature is significant!

1.5 Heisenberg's Uncertainty Principle (1927)

It is fundamentally impossible to simultaneously determine the **exact position and exact momentum** of a particle. The more precisely you know position, the less precisely you know momentum, and vice versa.

FORMULA

$$\Delta x \cdot \Delta p \geq h / 4\pi$$

where:

Δx = uncertainty in position (m)

Δp = uncertainty in momentum (kg·m/s)

h = Planck's constant

The product of uncertainties is always $\geq h/4\pi$ ($= \hbar/2$)

Key Points

- This is NOT a limitation of instruments — it is a fundamental property of nature
- Because we cannot fix an electron's exact position, we speak of **probability of finding electron** — this is the basis of quantum mechanics
- Classical orbit concept is replaced by **atomic orbital** (probability region)

Fundamentals of Quantum Mechanics

The postulates that govern the behavior of microscopic particles

3.1 Why Do We Need Quantum Mechanics?

Classical mechanics (Newton's laws) works perfectly for large objects (balls, planets). But at the scale of electrons and atoms, classical mechanics gives wrong predictions. **Quantum mechanics** was developed to correctly describe the behavior of microscopic particles.

3.2 Basic Postulates of Quantum Mechanics

Quantum mechanics is built on a set of fundamental assumptions (postulates) that cannot be derived — they are accepted because they correctly predict experimental observations.

Postulate 1: Wave Function (Ψ)

The state of a quantum system is completely described by a **wave function $\Psi(x, y, z, t)$** — a mathematical function of position and time. All measurable information about the particle is contained in Ψ .

Postulate 2: Physical Meaning of Ψ^2 (Born Interpretation)

The wave function itself has no direct physical meaning, but **$|\Psi|^2$ (or Ψ^2)** gives the probability density — the probability of finding the particle in a small volume element dV around a given point. Therefore:

FORMULA

$$\int |\Psi|^2 dV = 1 \text{ (Normalization Condition)}$$

The total probability of finding the particle somewhere in all space = 1 (certainty)

Postulate 3: Operators and Observables

Every measurable quantity (observable) in classical mechanics — such as energy, momentum, position — corresponds to a **quantum mechanical operator**. The measured value is obtained by applying the operator to the wave function.

Observable	Classical Symbol	QM Operator
Position (x)	x	$\hat{x}$ = multiply by x
Momentum (p)	mv	$\hat{p} = -i\hbar (\partial/\partial x)$
Kinetic Energy	$p^2/2m$	$\hat{T} = -(\hbar^2/2m)(\partial^2/\partial x^2)$
Total Energy	E	$\hat{H}$ (Hamiltonian operator)

Postulate 4: Eigenvalue Equation

When an operator $\hat{A}$ acts on a wave function Ψ and gives back the same Ψ multiplied by a constant 'a', then 'a' is a measurable value (eigenvalue) and Ψ is called an eigenfunction:

FORMULA

$$\hat{A} \Psi = a \Psi$$

e.g., $\hat{H} \Psi = E \Psi$ (Schrödinger equation is of this form)

Postulate 5: Superposition Principle

If Ψ_1 and Ψ_2 are valid wave functions, then any linear combination ($\Psi = c_1\Psi_1 + c_2\Psi_2$) is also a valid wave function. This is the basis of the LCAO method in MOT.

3.3 Requirements for a Valid Wave Function

Not every mathematical function can be a wave function. Ψ must satisfy these conditions:

Single-Valued

At any point in space, Ψ can have only one value (probability must be definite)

Continuous

Ψ and its first derivatives must be continuous

Normalizable

$\int |\Psi|^2 dV$ must be finite (total probability must be 1, not infinity)

everywhere (no sudden jumps)

Schrödinger Wave Equation

The fundamental equation of quantum mechanics — derived step by step

4.1 Need for the Schrödinger Equation

Just as Newton's $F = ma$ describes motion of classical particles, **Schrödinger's equation** describes the behavior of quantum particles (electrons, atoms). It tells us how the wave function changes with position and how to calculate allowed energy levels.

4.2 Derivation of Time-Independent Schrödinger Equation

1 Start with a Classical Wave

A particle moving in one dimension (x) can be represented as a wave: $\Psi(x) = A \sin(kx) + B \cos(kx)$, where $k = 2\pi/\lambda$ (wave number)

2 Use de Broglie Relation

From de Broglie: $\lambda = h/mv = h/p \rightarrow k = 2\pi/\lambda = 2\pi p/h = p/\hbar$, where $\hbar = h/2\pi$

3 Differentiate the Wave Function Twice

Differentiating $\Psi = A e^{i(kx)}$ twice: $d^2\Psi/dx^2 = -k^2\Psi = -(p^2/\hbar^2)\Psi$
Therefore: $p^2\Psi = -\hbar^2 (d^2\Psi/dx^2)$

4 Apply Total Energy ($E = KE + PE$)

Total energy $E = (p^2/2m) + V(x)$, where V = potential energy
So: $p^2 = 2m(E - V)$

5 Combine and Rearrange

Substituting p^2 into the differentiated equation:
 $-\hbar^2(d^2\Psi/dx^2) = 2m(E - V)\Psi$
Rearranging: $-(\hbar^2/2m)(d^2\Psi/dx^2) + V(x)\Psi = E\Psi$

$$-(\hbar^2/2m)(d^2\Psi/dx^2) + V(x)\Psi = E\Psi$$

$$\text{OR in 3D: } -(\hbar^2/2m)\nabla^2\Psi + V\cdot\Psi = E\Psi$$

$\hbar = h/2\pi = 1.055 \times 10^{-34}$ J-s (reduced Planck's constant)

m = mass of particle (kg)

Ψ = wave function

$V(x)$ = potential energy at position x

E = total energy of the particle

∇^2 = Laplacian operator = $\partial^2/\partial x^2 + \partial^2/\partial y^2 + \partial^2/\partial z^2$



Compact Form using Hamiltonian Operator ($\hat{H}$)

The left side of Schrödinger's equation is the Hamiltonian operator $\hat{H}$ acting on Ψ :

$$\hat{H}\Psi = E\Psi$$

where $\hat{H} = -(\hbar^2/2m)\nabla^2 + V$ is the total energy operator. Solving this equation gives allowed energies (E) and corresponding wave functions (Ψ).

4.3 Physical Meaning and Significance

Aspect	Explanation
What Ψ represents	A mathematical description of the quantum state; carries all info about the particle
Solving the equation	Gives discrete (quantized) values of E — energy levels of the system
Each solution	Gives one allowed energy E_n and its corresponding wave function Ψ_n
Why quantization occurs	Boundary conditions restrict which Ψ are valid $\rightarrow$ only certain E values are allowed
Reduces to classical for large n	Correspondence Principle — quantum result $\rightarrow$ classical result at large quantum numbers

Physical Meaning of Wave Function Ψ and Ψ^2

Interpreting what the wave function actually means physically

5.1 The Wave Function Ψ

The wave function Ψ (psi) is a mathematical function. It is obtained by solving Schrödinger's equation. By itself, Ψ has **no direct physical meaning** because it can be complex (involving $\sqrt{-1}$).

 **Think of It This Way**

Ψ is like a "recipe" or "map" — it encodes information about the particle's state, but to extract a physically meaningful quantity, you must use Ψ^2 .

5.2 Probability Density — Ψ^2 (or $|\Psi|^2$)

Max Born (1926) proposed the **Born Interpretation**: the square of the wave function gives the **probability density** — the probability of finding the particle per unit volume at a given point.

FORMULA

Probability $\propto |\Psi(x)|^2 dx$

$|\Psi(x)|^2 = \Psi^*(x) \cdot \Psi(x)$ (where Ψ^* is the complex conjugate of Ψ)
For real wave functions: $|\Psi|^2 = \Psi^2$
 $|\Psi|^2 dx$ = probability of finding particle between x and $x+dx$

5.3 Properties of Ψ and Ψ^2

Property	Ψ (Wave Function)	Ψ^2 (Probability Density)
Physical meaning	None directly — mathematical function	Probability of locating particle
Sign	Can be positive, negative, or complex	Always ≥ 0 (probability cannot be negative)
At node	$\Psi = 0$	$\Psi^2 = 0$ (zero probability)
Normalization	$\int \Psi^* \Psi dV = 1$	Total probability = 1

5.4 Diagrams: Wave Functions and Probability Density for Particle in Box

WAVE FUNCTIONS (Ψ) AND PROBABILITY DENSITIES (Ψ^2) FOR $N = 1, 2, 3$



Fig. 5.1 – Wave functions Ψ_n (top) and probability densities Ψ_n^2 (bottom) for $n=1,2,3$ in a 1D box. Note: number of nodes = $n-1$; number of Ψ^2 maxima = n .

🔑 Key Observations from the Diagrams

- Ψ can be positive or negative (like a wave), but Ψ^2 is always ≥ 0
- A **node** is where $\Psi = 0 \rightarrow$ no probability of finding particle there
- For quantum number n , there are **($n-1$) nodes**
- Higher $n \rightarrow$ more nodes $\rightarrow$ higher energy $\rightarrow$ more complex wave function
- Ψ^2 peaks show where the electron is most likely to be found

Particle in a One-Dimensional Box & Energy Quantization

The simplest quantum system — deriving energy levels from scratch

6.1 What is the Particle-in-a-Box Model?

Imagine a particle (like an electron) confined inside a **box** with perfectly rigid walls. The particle can move freely inside but cannot escape. This simple model demonstrates why energy is quantized in quantum mechanics.



Real-World Analogy

Think of a ball bouncing inside a box. Classically, the ball can have any energy. But quantum mechanically, only certain energies are allowed — like only certain harmonics are possible for a string fixed at both ends!

6.2 Model and Assumptions

PARTICLE IN A 1D BOX — PHYSICAL SETUP

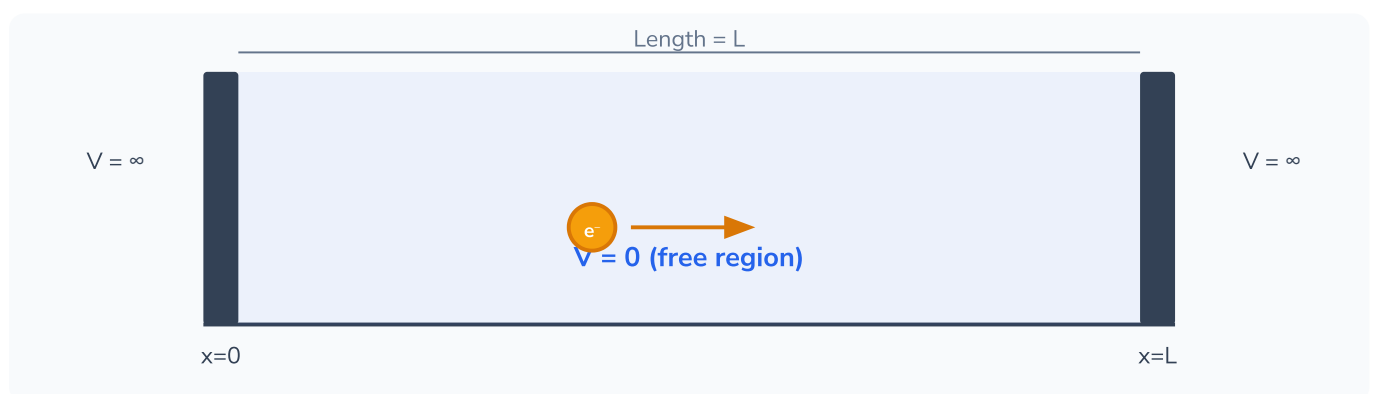


Fig. 6.1 – A particle (electron) confined in a 1D box of length L . $V = 0$ inside; $V = \infty$ at walls (particle cannot escape).

Assumptions of the Model

- The box has length L (from $x = 0$ to $x = L$)
- Potential energy $V = 0$ inside the box (particle moves freely)
- Potential energy $V = \infty$ at the walls and outside (particle cannot escape)
- The particle does not interact with the walls

- Motion is only in one dimension (x-direction)

6.3 Step-by-Step Derivation of Energy Levels

1 Write Schrödinger Equation Inside the Box ($V = 0$)

Since $V = 0$ inside: $-(\hbar^2/2m)(d^2\Psi/dx^2) = E\Psi$

Rearranging: $d^2\Psi/dx^2 = -(2mE/\hbar^2)\Psi$

Let $k^2 = 2mE/\hbar^2 \rightarrow d^2\Psi/dx^2 = -k^2\Psi$

2 General Solution

The general solution is: $\Psi(x) = A \sin(kx) + B \cos(kx)$

3 Apply Boundary Conditions

At $x = 0$: $\Psi(0) = 0$ (particle cannot be at wall)

$\rightarrow A \sin(0) + B \cos(0) = 0 \rightarrow 0 + B = 0 \rightarrow \mathbf{B = 0}$

So: $\Psi(x) = A \sin(kx)$

At $x = L$: $\Psi(L) = 0$

$\rightarrow A \sin(kL) = 0 \rightarrow \sin(kL) = 0$ (since $A \neq 0$)

$\rightarrow kL = n\pi$, where $n = 1, 2, 3, \dots$ ($n \neq 0$, otherwise $\Psi = 0$ everywhere)

Therefore: $\mathbf{k = n\pi/L}$

4 Find the Allowed Energies

From $k^2 = 2mE/\hbar^2$ and $k = n\pi/L$:

$(n\pi/L)^2 = 2mE_n/\hbar^2 \rightarrow E_n = n^2\pi^2\hbar^2/(2mL^2)$

Since $\hbar = h/2\pi$, we have $\hbar^2 = h^2/4\pi^2$, so:

$E_n = n^2\pi^2(h^2/4\pi^2)/(2mL^2) = \mathbf{n^2h^2/(8mL^2)}$

5 Normalize the Wave Function

Applying normalization ($\int_0^L |\Psi|^2 dx = 1$):

$\int_0^L A^2 \sin^2(n\pi x/L) dx = 1 \rightarrow A^2 \cdot L/2 = 1 \rightarrow A = \sqrt{2/L}$

Therefore: $\Psi_n(x) = \sqrt{2/L} \cdot \sin(n\pi x/L)$

FORMULA

$$E_n = n^2 h^2 / (8mL^2)$$

$$\Psi_n(x) = \sqrt{2/L} \cdot \sin(n\pi x/L)$$

E_n = energy of nth level (J)

n = quantum number = 1, 2, 3, ... (positive integers only; $n=0$ gives $\Psi=0$, not physical)

h = Planck's constant = 6.626×10^{-34} J·s

m = mass of particle (kg)
 L = length of the box (m)
 $\Psi_n(x)$ = normalized wave function for quantum state n

7.1 Energy Quantization — Key Observations

ENERGY LEVEL DIAGRAM — PARTICLE IN 1D BOX

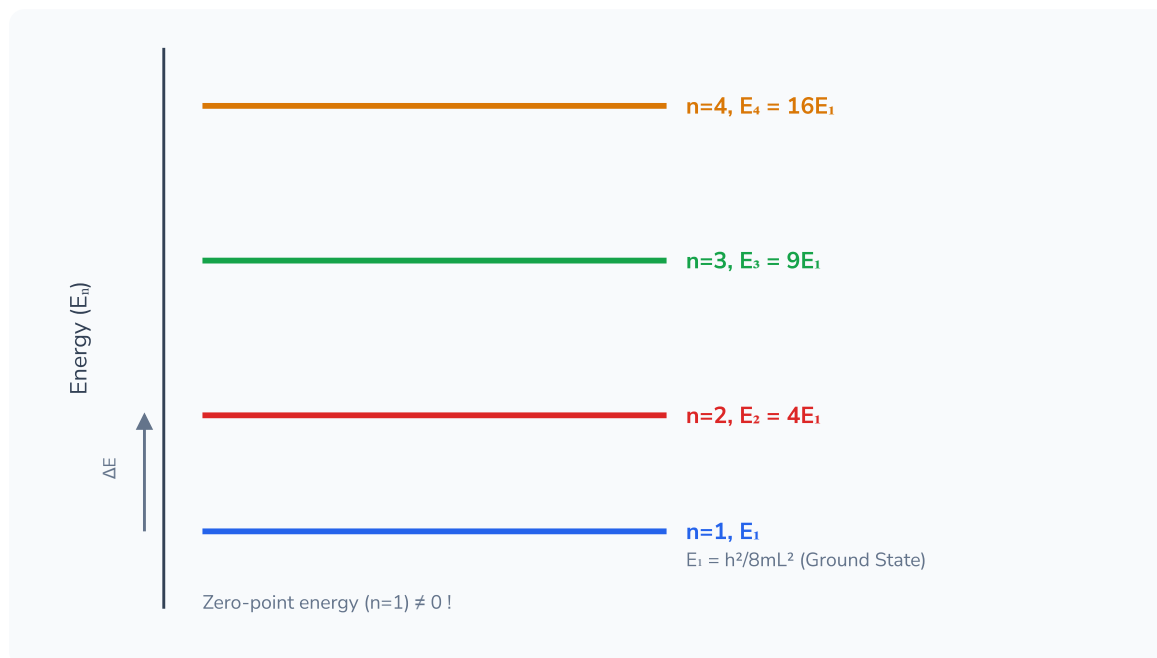


Fig. 7.1 – Quantized energy levels for particle in 1D box. E_1 is zero-point energy (particle always has some energy). $E_n \propto n^2$.

n	Energy E_n	Wave function nodes	Remarks
1	$E_1 = h^2/8mL^2$	0 (no nodes)	Ground state, lowest possible energy (Zero-Point Energy)
2	$E_2 = 4E_1$	1	First excited state
3	$E_3 = 9E_1$	2	Second excited state
4	$E_4 = 16E_1$	3	Third excited state
n	$E_n = n^2E_1$	$n-1$	General case

🔑 Critical Points About Zero-Point Energy

- **Zero-point energy (ZPE)** = $E_1 = h^2/8mL^2$. Even at absolute zero temperature, the particle has energy!
- This is a purely quantum effect — it arises from Heisenberg Uncertainty Principle (if $E=0$, position and momentum would be perfectly known → violates HUP)
- As L increases → E_1 decreases → for macroscopic boxes, levels are so close they appear continuous (correspondence principle)

- Smaller box (smaller L) $\rightarrow$ larger energy spacing $\rightarrow$ more pronounced quantum effects

Molecular Orbital Theory (MOT) & LCAO Method

How atomic orbitals combine to form molecular orbitals

8.1 Need for Molecular Orbital Theory

When atoms come together to form a molecule, electrons are no longer confined to individual atoms — they are shared by the entire molecule. We need a theory that describes electrons in **molecules**, not just in atoms.

Valence Bond Theory (VBT) — Older Approach

- Electrons remain in atomic orbitals
- Bond = overlap of two AOs
- Cannot explain paramagnetism of O_2
- Cannot explain delocalization in benzene
- Fails for many molecules

Molecular Orbital Theory (MOT) — Better

- Electrons belong to entire molecule
- AOs combine to form MOs spanning whole molecule
- Correctly explains O_2 paramagnetism
- Explains aromaticity, delocalization
- More powerful and accurate

8.2 Basic Concepts of MOT

1

Molecular Orbitals belong to the entire molecule

In a molecule, electrons occupy **molecular orbitals (MOs)** that extend over all nuclei. Just as electrons in atoms occupy atomic orbitals (s, p, d), electrons in molecules occupy MOs.

2

Two Types of MOs

Bonding MO (BMO): Lower energy than the original AOs. Electron density concentrated between nuclei → stabilizes the molecule. Symbol: σ , π

Antibonding MO (ABMO): Higher energy than original AOs. Electron density outside the internuclear region; has a nodal plane between nuclei → destabilizes molecule. Symbol: σ^* , π^*

3

Number of MOs = Number of AOs

When 'n' atomic orbitals combine, they produce exactly 'n' molecular orbitals. For two atoms each contributing one orbital: 1 BMO + 1 ABMO = 2 MOs total.

9.1 The LCAO Method (Linear Combination of Atomic Orbitals)

The mathematical method used to construct MOs from AOs is called **LCAO (Linear Combination of Atomic Orbitals)**. It is based on Postulate 5 (superposition principle).

1 The Basic Idea

If atom A has wave function Ψ_A and atom B has wave function Ψ_B , the molecular orbital Ψ_{MO} is formed by adding or subtracting these wave functions (linear combination).

FORMULA

$\Psi_{\text{Bonding}} = \Psi_A + \Psi_B$ (constructive combination → lower energy)

$\Psi_{\text{Antibonding}} = \Psi_A - \Psi_B$ (destructive combination → higher energy)

Ψ_A = wave function of atomic orbital on atom A

Ψ_B = wave function of atomic orbital on atom B

Addition → increased electron density between nuclei (bonding)

Subtraction → decreased electron density between nuclei (antibonding, with node)

FORMATION OF BONDING AND ANTIBONDING MOS FROM 1S ATOMIC ORBITALS

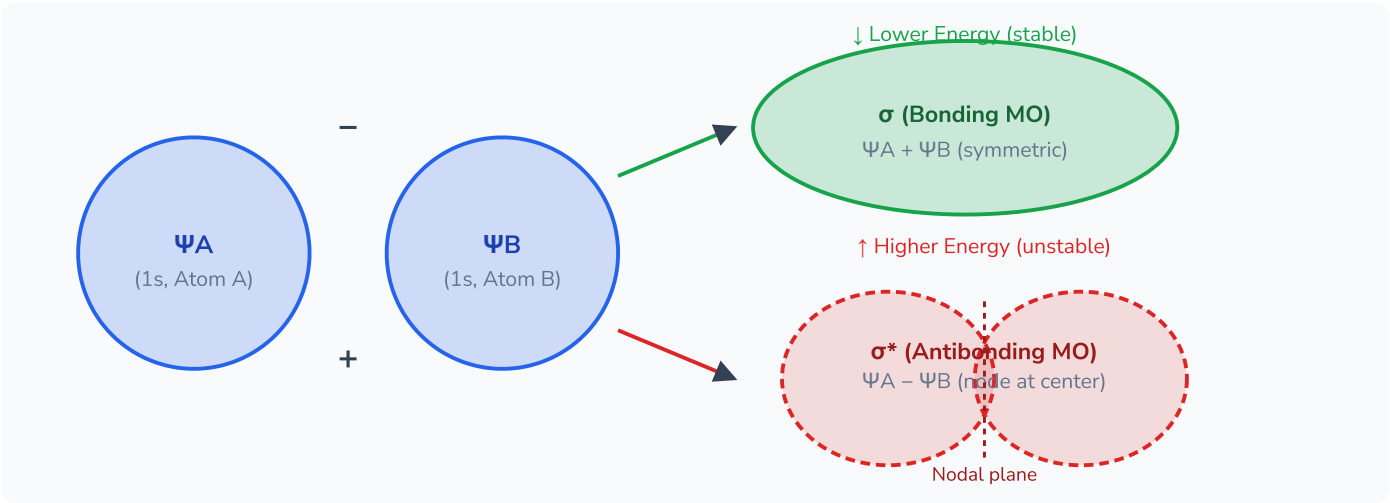


Fig. 9.1 – LCAO: Two 1s AOs combine to form σ (bonding, lower energy) and σ^* (antibonding, higher energy, with nodal plane).
The bonding MO has greater electron density between nuclei.

9.2 Conditions for Effective LCAO Combination

Condition	Explanation
1. Similar Energy	Combining AOs must have similar (comparable) energies. Large energy difference → poor combination
2. Similar Size (Overlap)	Orbitals must have significant spatial overlap. Minimal overlap → weak bonding

3. Symmetry Match

Orbitals must have compatible symmetry. Mismatched symmetry → zero overlap → no combination

9.3 Types of Molecular Orbitals Formed

AO Combination	MOs Formed	Character
s + s (head-on)	σ_{1s} , σ^*_{1s}	sigma (σ) — symmetric about bond axis
p + p (head-on)	σ_{2p} , σ^*_{2p}	sigma (σ)
p + p (side-on)	π_{2p} , π^*_{2p}	pi (π) — two lobes, one above/below axis
s + p (head-on)	σ (mixed), σ^* (mixed)	sigma — occurs in heteronuclear molecules

Bonding in Homonuclear & Heteronuclear Diatomic Molecules

Filling molecular orbitals according to Aufbau, Hund's and Pauli's rules

10.1 Rules for Filling Molecular Orbitals

Electrons fill MOs by the same rules used for atoms:

Aufbau Principle

Fill MOs in order of increasing energy (lowest energy first)

Pauli's Exclusion

Each MO can hold a maximum of 2 electrons with opposite spins

Hund's Rule

Among degenerate MOs (same energy), fill one electron each before pairing

10.2 MO Energy Order for Homonuclear Diatomics

FORMULA

For $Z \leq 7$ (Li_2 to N_2):

$$\sigma_{1s} < \sigma^*_{1s} < \sigma_{2s} < \sigma^*_{2s} < \pi_{2p} = \pi_{2p} < \sigma_{2p} < \pi^*_{2p} = \pi^*_{2p} < \sigma^*_{2p}$$

For $Z \geq 8$ (O_2 , F_2):

$$\sigma_{1s} < \sigma^*_{1s} < \sigma_{2s} < \sigma^*_{2s} < \sigma_{2p} < \pi_{2p} = \pi_{2p} < \pi^*_{2p} = \pi^*_{2p} < \sigma^*_{2p}$$

Note: The order of σ_{2p} and π_{2p} swaps between N_2 and O_2 — this is a well-established experimental observation

10.3 Homonuclear Molecules — Electronic Configurations

Molecule	Total e ⁻	Configuration (abbreviated)	Bond Order	Magnetic Property
H ₂	2	(σ_{1s}) ²	1	Diamagnetic
He ₂	4	(σ_{1s}) ² (σ^*_{1s}) ²	0	Does not exist

Li_2	6	$\dots(\sigma 2s)^2$	1	Diamagnetic
Be_2	8	$\dots(\sigma 2s)^2(\sigma^* 2s)^2$	0	Does not exist
B_2	10	$\dots(\pi 2p)^1(\pi 2p)^1$	1	Paramagnetic
C_2	12	$\dots(\pi 2p)^2(\pi 2p)^2$	2	Diamagnetic
N_2	14	$\dots(\sigma 2p)^2(\pi 2p)^2(\pi 2p)^2$	3	Diamagnetic
O_2	16	$\dots(\sigma 2p)^2(\pi 2p)^2(\pi 2p)^2(\pi^* 2p)^1(\pi^* 2p)^1$	2	Paramagnetic
F_2	18	$\dots(\pi^* 2p)^2(\pi^* 2p)^2$	1	Diamagnetic
Ne_2	20	$\dots(\sigma^* 2p)^2$	0	Does not exist



O₂ Paramagnetism — A Great Success of MOT

VBT predicts O₂ has all paired electrons (diamagnetic). But experiments show O₂ is attracted to magnets (paramagnetic → has unpaired electrons). MOT correctly predicts 2 unpaired electrons in degenerate $\pi^* 2p$ orbitals — one of MOT's greatest triumphs!

11.1 Heteronuclear Diatomic Molecules — CO

In heteronuclear molecules (atoms $A \neq B$), the AOs of different atoms have different energies. The more electronegative atom has lower-energy orbitals. Despite this, CO follows a MO pattern similar to N₂ (isoelectronic: same number of electrons = 14).

CO Molecule — Key Facts

- Carbon: 6 electrons; Oxygen: 8 electrons → Total = 14 electrons (isoelectronic with N₂)
- Bond Order = 3 (triple bond, like N₂)
- Diamagnetic (all electrons paired)
- Highest occupied MO (HOMO) = $\sigma 2p$ (slightly antibonding contribution from C end)
- CO is a Lewis base (donates electron pair through HOMO) — important in coordination chemistry

Energy Level Diagrams — O₂ and CO Molecules

Visual representation of MO filling with detailed energy levels

12.1 Energy Level Diagram of O₂

MOLECULAR ORBITAL ENERGY LEVEL DIAGRAM — O₂ MOLECULE

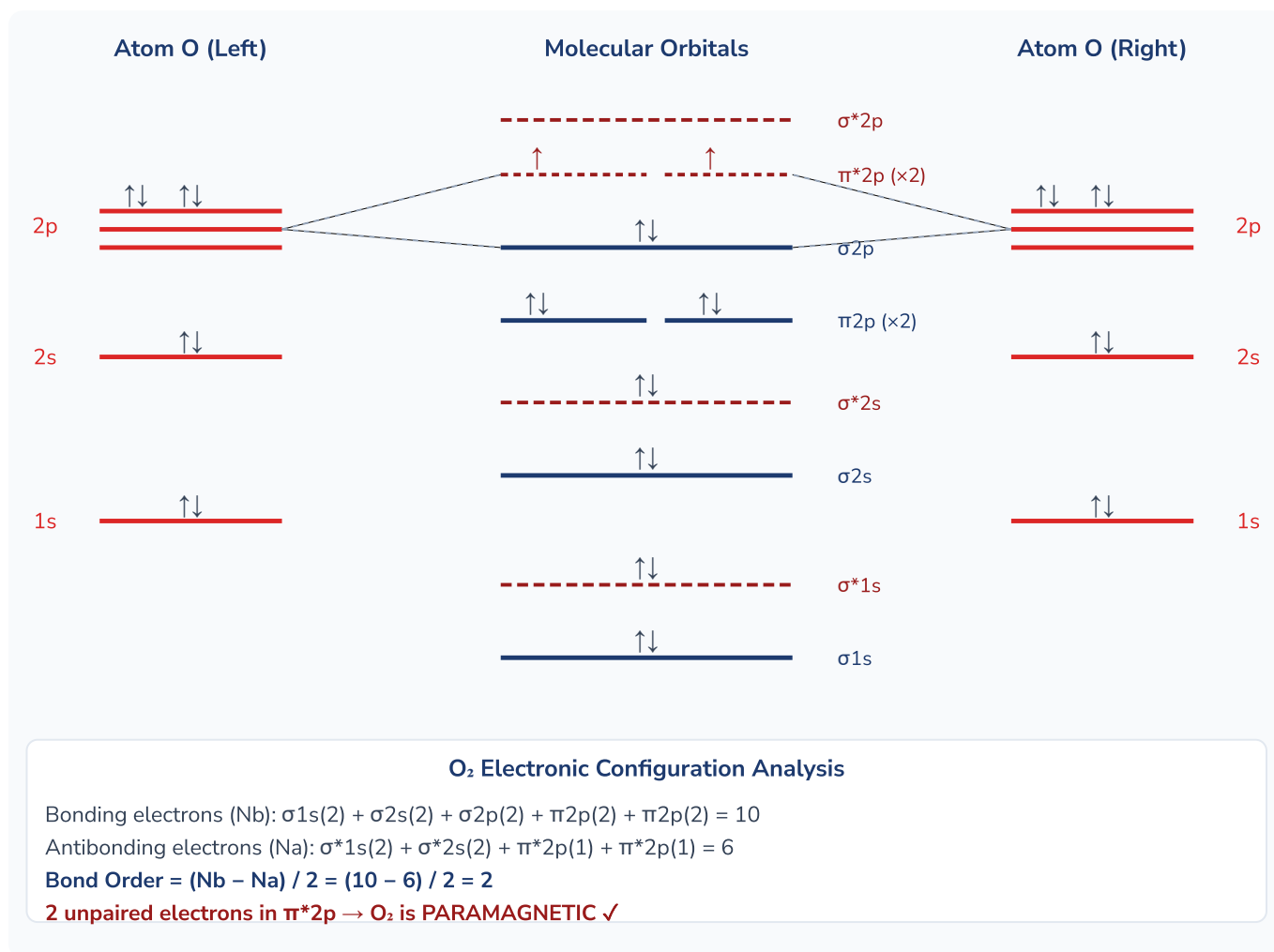
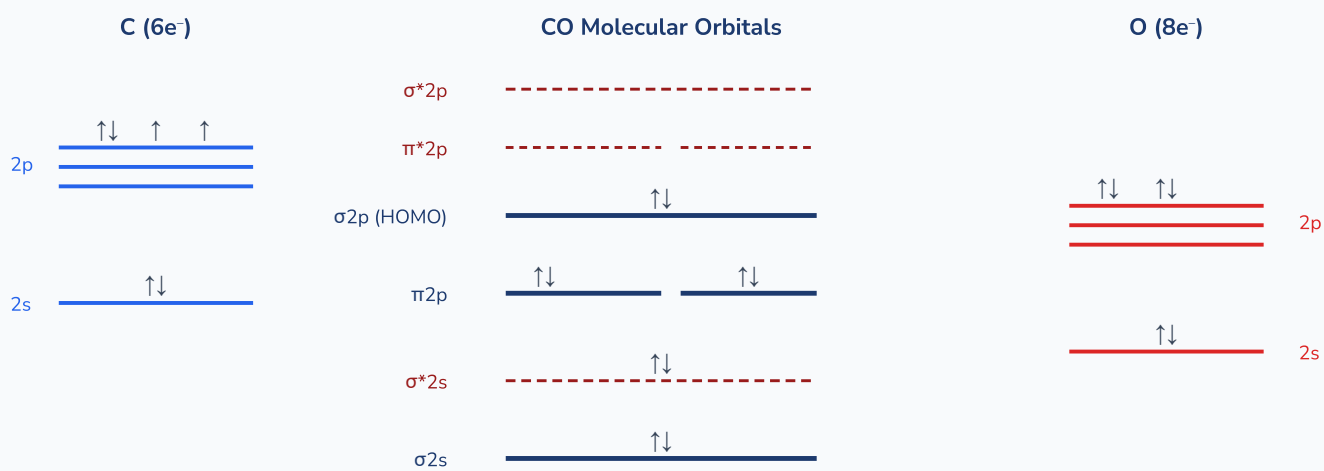


Fig. 12.1 – Complete MO energy level diagram for O₂ (16 electrons). Note the 2 unpaired electrons in the degenerate $\pi^* 2p$ orbitals, explaining the paramagnetic behavior of O₂.

12.2 Energy Level Diagram of CO

MOLECULAR ORBITAL ENERGY LEVEL DIAGRAM — CO MOLECULE



CO Analysis (14 electrons, isoelectronic with N₂)

Bonding (Nb): $\sigma_{1s}(2) + \sigma_{2s}(2) + \pi_{2p}(4) + \sigma_{2p}(2) = 10$

Antibonding (Na): $\sigma^*_{1s}(2) + \sigma^*_{2s}(2) = 4$

Bond Order = $(10 - 4)/2 = 3$ | Diamagnetic (all paired) ✓

Fig. 12.2 – MO energy level diagram for CO. Note oxygen AOs are at lower energy (more electronegative). CO has bond order = 3 and is diamagnetic. HOMO is the σ_{2p} orbital.

π -Molecular Orbitals — Butadiene and Benzene

Delocalized pi systems and their energy levels using Hückel theory

13.1 What are π -Molecular Orbitals?

In conjugated molecules (alternating single and double bonds), the p orbitals on each carbon are perpendicular to the molecular plane. These p orbitals overlap sideways to form a **delocalized π system** extending over the entire molecule. This is described by π -molecular orbitals.



Key Insight

Delocalized π electrons are NOT confined between two atoms — they are shared over all the carbon atoms in the conjugated system. This delocalization lowers the overall energy (stabilizes the molecule) — the concept of

resonance stabilization

or

aromatic stabilization

.

13.2 π -MOs in Butadiene ($\text{CH}_2=\text{CH}-\text{CH}=\text{CH}_2$)

Butadiene has 4 carbon atoms, each contributing one p orbital. By LCAO, 4 p orbitals form **4 π -molecular orbitals** ($\psi_1, \psi_2, \psi_3, \psi_4$) with increasing energy and increasing number of nodes.

π -MOLECULAR ORBITALS OF BUTADIENE (4 CARBON ATOMS, 4 π ELECTRONS)

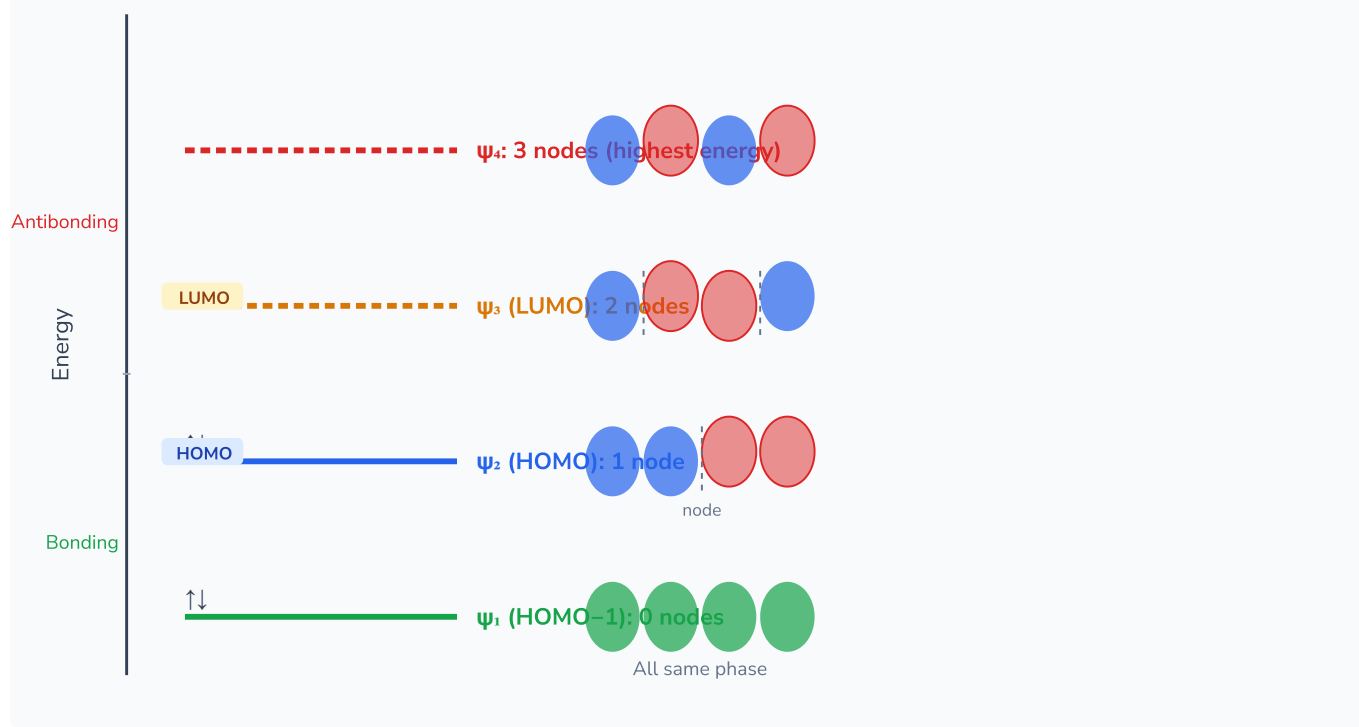


Fig. 13.1 – π -MOs of butadiene. 4 electrons fill ψ_1 and ψ_2 (both bonding MOs). HOMO = ψ_2 (1 node), LUMO = ψ_3 (2 nodes). Blue = positive phase, Red = negative phase of wave function.

13.3 π -MOs in Benzene (C_6H_6)

Benzene has 6 carbon atoms in a ring, each contributing one p orbital. These form **6 π -molecular orbitals** arranged in 3 pairs. Benzene has 6 π electrons that fill the 3 bonding MOs.

π -MOLECULAR ORBITALS OF BENZENE — ENERGY LEVEL DIAGRAM

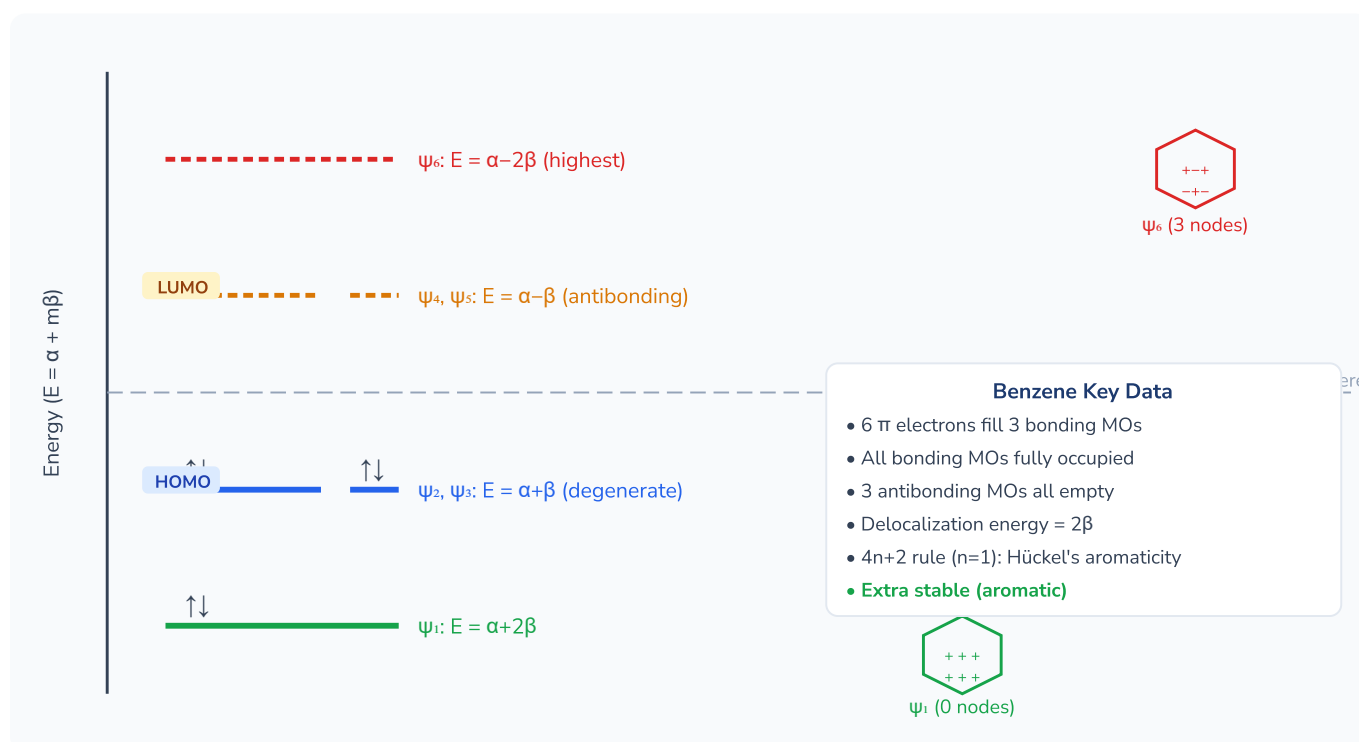


Fig. 13.2 – π -MO energy levels of benzene. 6 electrons fill 3 bonding MOs (ψ_1 , ψ_2 , ψ_3). All antibonding MOs are empty $\rightarrow$ benzene is exceptionally stable (aromatic). α = coulomb integral, β = resonance integral (negative value $\rightarrow \beta$ makes bonding stable).

13.4 Aromaticity — Hückel's Rule

FORMULA

Aromatic if: number of π electrons = $4n + 2$, where $n = 0, 1, 2, 3, \dots$

$n=0$: 2 π electrons (cyclopropene cation)

$n=1$: 6 π electrons (benzene ✓, cyclopentadienyl anion ✓)

$n=2$: 10 π electrons (naphthalene, azulene)

Anti-aromatic: $4n$ electrons (cyclobutadiene, $n=1$: 4 electrons)

Non-aromatic: not cyclic or not conjugated

Bond Order and Molecular Stability

Calculating and interpreting bond order from molecular orbital configurations

14.1 Definition of Bond Order

Bond order is a number that tells us the **effective number of chemical bonds** between two atoms. It is calculated from the MO configuration by counting electrons in bonding vs antibonding orbitals.

FORMULA

$$\text{Bond Order (B.O.)} = (\text{Nb} - \text{Na}) / 2$$

Nb = number of electrons in bonding molecular orbitals

Na = number of electrons in antibonding molecular orbitals

The factor of 2 accounts for the fact that each bond is formed by 2 electrons

Bond order can be 0, 0.5, 1, 1.5, 2, 2.5, 3, 3.5 (including fractional values!)

14.2 Significance of Bond Order

Bond Order	Meaning	Bond Strength	Bond Length	Example
0	No bond; molecule does not exist	Zero	—	He ₂ , Ne ₂
0.5	Very weak bond	Very weak	Very long	H ₂ ⁺
1	Single bond	Weak	Long	F ₂ , H ₂ , Li ₂
1.5	Bond between single and double	Medium	Medium	NO, O ₂ ⁻
2	Double bond	Strong	Short	O ₂ , C ₂
2.5	Bond between double and triple	Very strong	Very short	NO ⁺
3	Triple bond	Strongest	Shortest	N ₂ , CO

🔑 Bond Order Relationships

- Higher bond order → Stronger bond → Larger bond dissociation energy
- Higher bond order → Shorter bond length

- Bond order = 0 $\rightarrow$ molecule does not exist in stable form
- Fractional bond orders (0.5, 1.5, 2.5) are perfectly valid in MOT!
- Paramagnetic if MO configuration has any unpaired electrons
- Diamagnetic if all electrons in MOs are paired

14.3 Bond Order Calculations — Summary Table

Species	Total e ⁻	Nb	Na	Bond Order	Exists?	Magnetic
H ₂ ⁺	1	1	0	0.5	Yes (unstable)	Para
H ₂	2	2	0	1	Yes	Dia
He ₂ ⁺	3	2	1	0.5	Yes (barely)	Para
He ₂	4	2	2	0	No	—
N ₂	14	10	4	3	Yes	Dia
O ₂	16	10	6	2	Yes	Para
O ₂ ⁺	15	10	5	2.5	Yes	Para
O ₂ ⁻	17	10	7	1.5	Yes	Para
O ₂ ²⁻	18	10	8	1	Yes (peroxide)	Dia
F ₂	18	10	8	1	Yes	Dia
Ne ₂	20	10	10	0	No	—
CO	14	10	4	3	Yes	Dia
NO	15	10	5	2.5	Yes	Para

Solved Numerical Problems (40+)

Step-by-step solutions for all types of exam problems

Part A: Particle in 1D Box Problems

Problem 1 — Ground State Energy of Electron in a Box

Q: Calculate the ground state energy of an electron confined in a 1D box of length 1 nm (1×10^{-9} m).

Given:

$n = 1$ (ground state), $L = 1 \times 10^{-9}$ m

$h = 6.626 \times 10^{-34}$ J·s, $m = 9.109 \times 10^{-31}$ kg

Solution:

Using: $E_n = n^2 h^2 / (8mL^2)$

$$E_1 = (1)^2 \times (6.626 \times 10^{-34})^2 / [8 \times (9.109 \times 10^{-31}) \times (1 \times 10^{-9})^2]$$

$$E_1 = (4.390 \times 10^{-67}) / [8 \times 9.109 \times 10^{-31} \times 10^{-18}]$$

$$E_1 = (4.390 \times 10^{-67}) / (7.287 \times 10^{-48})$$

$$E_1 = 6.025 \times 10^{-20} \text{ J}$$

✓ **Answer:** $E_1 = 6.025 \times 10^{-20} \text{ J} \approx 0.376 \text{ eV}$

Problem 2 — Energy of $n=3$ Level

Q: For the same box ($L = 1$ nm), find the energy of the $n=3$ state and the transition energy from $n=3$ to $n=1$.

Solution:

$$E_3 = n^2 \times E_1 = 9 \times 6.025 \times 10^{-20} = 5.423 \times 10^{-19} \text{ J}$$

$$E_1 = 6.025 \times 10^{-20} \text{ J (from Problem 1)}$$

$$\Delta E (3 \rightarrow 1) = E_3 - E_1 = 5.423 \times 10^{-19} - 6.025 \times 10^{-20}$$

$$\Delta E = 5.423 \times 10^{-19} - 0.6025 \times 10^{-19} = 4.820 \times 10^{-19} \text{ J}$$

✓ **$E_3 = 5.423 \times 10^{-19} \text{ J}$ | $\Delta E_{3 \rightarrow 1} = 4.820 \times 10^{-19} \text{ J}$**

Problem 3 — de Broglie Wavelength of Electron

Q: Calculate the de Broglie wavelength of an electron moving with velocity 2×10^6 m/s.

Given: $v = 2 \times 10^6$ m/s, $m = 9.109 \times 10^{-31}$ kg, $h = 6.626 \times 10^{-34}$ J·s

Solution:

$$\lambda = h/(mv) = (6.626 \times 10^{-34}) / (9.109 \times 10^{-31} \times 2 \times 10^6)$$

$$\lambda = (6.626 \times 10^{-34}) / (1.822 \times 10^{-24})$$

$$\lambda = 3.637 \times 10^{-10} \text{ m} = 0.3637 \text{ nm}$$

✓ $\lambda = 3.64 \times 10^{-10} \text{ m} = 3.64 \text{ \AA}$ (in X-ray range, confirming wave nature!)

Problem 4 — de Broglie Wavelength of Ball

Q: Calculate the de Broglie wavelength of a cricket ball of mass 0.1 kg moving at 30 m/s. Comment on the result.

Solution:

$$\lambda = h/(mv) = (6.626 \times 10^{-34}) / (0.1 \times 30)$$

$$\lambda = (6.626 \times 10^{-34}) / 3 = 2.21 \times 10^{-34} \text{ m}$$

✓ $\lambda = 2.21 \times 10^{-34} \text{ m}$ (immeasurably small — classical behavior valid!)

Problem 5 — Heisenberg Uncertainty Principle

Q: An electron is located with uncertainty in position $\Delta x = 0.1$ nm. What is the minimum uncertainty in its velocity?

Given: $\Delta x = 0.1 \times 10^{-9}$ m, $m = 9.109 \times 10^{-31}$ kg

Solution:

From HUP: $\Delta x \cdot \Delta p \geq h/4\pi$

$$\Delta p \geq h/(4\pi \cdot \Delta x) = (6.626 \times 10^{-34}) / (4\pi \times 0.1 \times 10^{-9})$$

$$\Delta p \geq (6.626 \times 10^{-34}) / (1.257 \times 10^{-9}) = 5.27 \times 10^{-25} \text{ kg·m/s}$$

$$\Delta v = \Delta p/m = (5.27 \times 10^{-25}) / (9.109 \times 10^{-31}) = 5.79 \times 10^5 \text{ m/s}$$

✓ $\Delta v_{\text{min}} \approx 5.79 \times 10^5 \text{ m/s}$ (enormous! Electron cannot be precisely located)

Problem 6 — Energy Level Ratio

Q: If the ground state energy of a particle in a box is 1.5 eV, what is the energy of the 4th excited state?

Solution:

Ground state = $n=1$, $E_1 = 1.5$ eV

4th excited state = $n = 5$ (since 1st excited = $n=2$, 2nd = $n=3$, 3rd = $n=4$, 4th = $n=5$)

$$E_5 = 5^2 \times E_1 = 25 \times 1.5 = 37.5 \text{ eV}$$

✓ Energy of 4th excited state ($n=5$) = 37.5 eV

Problem 7 — Box Length from Energy

Q: The ground state energy of an electron in a 1D box is 1.0×10^{-18} J. Calculate the length of the box.

Solution:

$$E_1 = h^2/(8mL^2) \rightarrow L^2 = h^2/(8mE_1)$$

$$L^2 = (6.626 \times 10^{-34})^2 / (8 \times 9.109 \times 10^{-31} \times 1.0 \times 10^{-18})$$

$$L^2 = (4.390 \times 10^{-67}) / (7.287 \times 10^{-48}) = 6.025 \times 10^{-20} \text{ m}^2$$

$$L = \sqrt{(6.025 \times 10^{-20})} = 2.45 \times 10^{-10} \text{ m}$$

✓ $L = 2.45 \times 10^{-10} \text{ m} = 2.45 \text{ \AA}$

Problem 8 — Wavelength of Absorbed Photon

Q: An electron in a 1D box of length 2 nm jumps from $n=1$ to $n=3$. Calculate the wavelength of photon absorbed.

Solution:

$$E_1 = h^2/(8mL^2) = (6.626 \times 10^{-34})^2 / (8 \times 9.109 \times 10^{-31} \times (2 \times 10^{-9})^2)$$

$$E_1 = 4.390 \times 10^{-67} / (2.915 \times 10^{-47}) = 1.506 \times 10^{-20} \text{ J}$$

$$\Delta E = E_3 - E_1 = (9-1) \times E_1 = 8 \times 1.506 \times 10^{-20} = 1.205 \times 10^{-19} \text{ J}$$

$$\lambda = hc/\Delta E = (6.626 \times 10^{-34} \times 3 \times 10^8) / (1.205 \times 10^{-19})$$

$$\lambda = 1.650 \times 10^{-6} \text{ m} = 1650 \text{ nm (infrared region)}$$

✓ $\lambda = 1650 \text{ nm}$ (infrared light is absorbed for this transition)

Part B: Bond Order Calculations

Problem 9 — Bond Order of N₂

Q: Write the MO configuration of N₂ and calculate its bond order.

Solution:

N₂ has 14 electrons total (7 from each N).

MO Configuration: $(\sigma 1s)^2(\sigma^* 1s)^2(\sigma 2s)^2(\sigma^* 2s)^2(\pi 2p)^2(\pi 2p)^2(\sigma 2p)^2$

[Note: For N₂, $\pi 2p$ fills before $\sigma 2p$]

Bonding electrons (Nb) = $\sigma 1s(2) + \sigma 2s(2) + \pi 2p(2) + \pi 2p(2) + \sigma 2p(2) = 10$

Antibonding electrons (Na) = $\sigma^* 1s(2) + \sigma^* 2s(2) = 4$

Bond Order = $(Nb - Na)/2 = (10 - 4)/2 = 6/2 = 3$

✓ N₂ Bond Order = 3 (triple bond) | All electrons paired → Diamagnetic

Problem 10 — Bond Order of O₂

Q: Calculate bond order of O₂ and determine its magnetic character.

Solution:

O₂ has 16 electrons. MO Configuration:

$(\sigma 1s)^2(\sigma^* 1s)^2(\sigma 2s)^2(\sigma^* 2s)^2(\sigma 2p)^2(\pi 2p)^2(\pi 2p)^2(\pi^* 2p)^1(\pi^* 2p)^1$

[For O₂, $\sigma 2p$ fills before $\pi^* 2p$]

Nb = 2+2+2+2+2 = 10; Na = 2+2+1+1 = 6

Bond Order = $(10-6)/2 = 4/2 = 2$

Unpaired electrons: 2 (one each in the two degenerate $\pi^* 2p$ orbitals)

✓ O₂: Bond Order = 2 | Paramagnetic (2 unpaired electrons)

Problem 11 — Bond Order of O₂⁺

Q: The O₂⁺ ion is formed by removing one electron from O₂. Find its bond order.

Solution:

O₂⁺ has 15 electrons (one removed from highest occupied MO = $\pi^* 2p$ of O₂).

Configuration: $\dots(\pi^* 2p)^1(\pi^* 2p)^0$ [one electron removed from $\pi^* 2p$]

Nb = 10 (same as O₂); Na = 6 - 1 = 5

Bond Order = $(10-5)/2 = 5/2 = 2.5$

Bond order 2.5 > 2 → O₂⁺ has stronger bond than O₂ → shorter bond length

✓ O₂⁺: Bond Order = 2.5 | Paramagnetic | Stronger bond than O₂

Problem 12 — Bond Order of F₂

Q: Write MO configuration of F₂ and find bond order.

Solution:

F₂ has 18 electrons.

Config: $(\sigma 1s)^2(\sigma^* 1s)^2(\sigma 2s)^2(\sigma^* 2s)^2(\sigma 2p)^2(\pi 2p)^2(\pi^* 2p)^2(\pi^* 2p)^2$

Nb = 2+2+2+2+2 = 10; Na = 2+2+2+2 = 8

Bond Order = $(10-8)/2 = 2/2 = 1$

✓ F₂: Bond Order = 1 (single bond) | All paired → Diamagnetic

Problem 13 — Bond Order of H₂

Q: Find bond order of H₂ and H₂⁺.

H₂ (2 electrons):

Config: $(\sigma 1s)^2$ | Nb=2, Na=0 | B.O. = $(2-0)/2 = 1$

H₂⁺ (1 electron):

Config: $(\sigma 1s)^1$ | Nb=1, Na=0 | B.O. = $(1-0)/2 = 0.5$

✓ H₂: B.O.=1 (exists) | H₂⁺: B.O.=0.5 (exists but weakly bonded)

Problem 14 — He₂ does NOT exist

Q: Predict stability of He₂ using MOT.

He₂ (4 electrons):

Config: $(\sigma 1s)^2(\sigma^* 1s)^2$ | Nb=2, Na=2

Bond Order = $(2-2)/2 = 0$

✓ He₂: B.O.=0 → He₂ does NOT exist. Bonding and antibonding effects cancel out completely.

Problem 15 — NO Molecule

Q: Calculate bond order of NO and find if it is paramagnetic.

NO has 15 electrons (N=7, O=8):

Config: $(\sigma 1s)^2(\sigma^* 1s)^2(\sigma 2s)^2(\sigma^* 2s)^2(\sigma 2p)^2(\pi 2p)^2(\pi^* 2p)^1$

Nb=10, Na=5

Bond Order = $(10-5)/2 = 2.5$

Unpaired e^- : 1 (in π^*2p)

✓ NO: B.O.=2.5 | Paramagnetic (1 unpaired electron)

More Bond Order & Energy Problems

Problem 16 — Energy Difference Between Levels

Q: $E_2 - E_1$ for electron in 0.5 nm box?

$E_1 = h^2/8mL^2$ with $L=0.5 \times 10^{-9}$; $E_1 = 2.41 \times 10^{-19}$ J; $\Delta E = E_2 - E_1 = (4-1) \times E_1 = 3 \times 2.41 \times 10^{-19} = 7.23 \times 10^{-19}$ J

✓ $\Delta E = 7.23 \times 10^{-19}$ J = 4.52 eV

Problem 17 — Energy of $n=2$ in 2 nm box

Q: Find E_2 for electron in 2 nm box.

$E_1 = 1.506 \times 10^{-20}$ J (computed earlier); $E_2 = 4 \times E_1 = 4 \times 1.506 \times 10^{-20} = 6.02 \times 10^{-20}$ J

✓ $E_2 = 6.02 \times 10^{-20}$ J

Problem 18 — Bond Order of C_2

Q: Calculate bond order of C_2 (12 electrons).

Config: $(\sigma 1s)^2(\sigma^* 1s)^2(\sigma 2s)^2(\sigma^* 2s)^2(\pi 2p)^2(\pi 2p)^2$; Nb=10, Na=4; B.O. = $(10-4)/2=3$? No — recheck:
Nb=2+2+2+2=8, Na=2+2=4; B.O. = $(8-4)/2=2$

✓ C_2 : B.O.=2 (double bond) | Diamagnetic

Problem 19 — Li_2 Bond Order

Q: Li_2 has 6 electrons. Find bond order.

Config: $(\sigma 1s)^2(\sigma^* 1s)^2(\sigma 2s)^2$; Nb=4, Na=2; B.O.= $(4-2)/2=1$

✓ Li_2 : B.O.=1 | Diamagnetic | Li_2 exists

Problem 20 — Photon Energy from Hydrogen Spectrum

Q: Calculate the energy of a photon with wavelength 656 nm (H α line).

$$E = hc/\lambda = (6.626 \times 10^{-34} \times 3 \times 10^8) / (656 \times 10^{-9}) = 1.988 \times 10^{-25} / 6.56 \times 10^{-7} = 3.03 \times 10^{-19} \text{ J}$$

✓ $E = 3.03 \times 10^{-19} \text{ J} = 1.89 \text{ eV}$

Problem 21 — Number of Nodes in Ψ_4

Q: How many nodes does the wave function Ψ_4 ($n=4$) of a particle in a 1D box have?

$$\text{Number of nodes} = n - 1 = 4 - 1 = 3$$

✓ Ψ_4 has 3 nodes (where $\Psi = 0$, between $x=0$ and $x=L$)

Problem 22 — ZPE Ratio in Different Boxes

Q: Compare zero-point energies of an electron in boxes of length L and $2L$.

$$E_1(L) = h^2/8mL^2; E_1(2L) = h^2/8m(2L)^2 = h^2/32mL^2; \text{Ratio} = E_1(L)/E_1(2L) = (h^2/8mL^2)/(h^2/32mL^2) = 32/8 = 4$$

✓ ZPE in box L is 4× larger than in box $2L$. Larger box $\rightarrow$ lower zero-point energy.

Problem 23 — MO Config of B_2

Q: Write MO config of B_2 (10 electrons) and find its magnetic character.

Config: $(\sigma 1s)^2(\sigma^* 1s)^2(\sigma 2s)^2(\sigma^* 2s)^2(\pi 2p)^1(\pi 2p)^1$ [Hund's rule: one in each degenerate $\pi 2p$]; Nb=6, Na=4; B.O.= $(6-4)/2=1$; Two unpaired electrons in degenerate $\pi 2p$

✓ B₂: B.O.=1 | Paramagnetic (2 unpaired e⁻)

Problem 24 — Finding velocity from de Broglie wavelength

Q: A particle has de Broglie wavelength 2.5×10^{-10} m. If mass = 6.64×10^{-27} kg (helium atom), find velocity.

$$\lambda = h/mv \rightarrow v = h/(m\lambda) = (6.626 \times 10^{-34}) / (6.64 \times 10^{-27} \times 2.5 \times 10^{-10}) = 6.626 \times 10^{-34} / (1.66 \times 10^{-36}) = 399 \text{ m/s}$$

✓ $v = 399 \text{ m/s} \approx 400 \text{ m/s}$ (thermal speed of He at room temperature!)

Problem 25 — Comparing Bond Lengths of N₂, O₂, F₂

Q: Arrange N₂, O₂, F₂ in increasing order of bond length using bond order.

Bond orders: N₂=3, O₂=2, F₂=1; Higher bond order → shorter bond; Order of bond length: N₂ < O₂ < F₂

✓ Bond length order: N₂ < O₂ < F₂ (N₂ has shortest, strongest bond; F₂ has longest, weakest)

Problems 26-35 — Rapid Fire Calculations

26. Bond order of Ne₂: Config ends in $(\sigma^*2p)^2$. Nb=10, Na=10. B.O.=0. Ne₂ does NOT exist.

27. NO⁺ (14 electrons): Config like CO. B.O.=(10-4)/2=3. Diamagnetic. Stronger bond than NO.

28. NO⁻ (16 electrons): Like O₂. B.O.=(10-6)/2=2. Paramagnetic.

29. O₂²⁻ (18 electrons, peroxide): Nb=10, Na=8. B.O.=(10-8)/2=1. Diamagnetic. This is the peroxide ion!

30. Particle in box: If E₃=54 eV, find E₁. $E_1 = E_3/n^2 = 54/9 = 6 \text{ eV}$.

31. π electrons in benzene for aromaticity check: $6 = 4(1)+2$, n=1. Aromatic ✓

32. Butadiene π electrons: 4. Number of π-MOs: 4. Filled MOs: ψ_1, ψ_2 (HOMO). First π transition: $\psi_2 \rightarrow \psi_3$

33. For n=5 in box, kinetic energy = $25 \times E_1 = 25 \times (h^2/8mL^2)$. Spacing between n=5 and n=4: $(25-16) \times E_1 = 9E_1$

34. He₂⁺ (3 electrons): $(\sigma 1s)^2(\sigma^* 1s)^1$. Nb=2, Na=1. B.O.=0.5. Exists weakly. Paramagnetic.

35. Photon frequency for E = 4.5×10^{-19} J: $\nu = E/h = 4.5 \times 10^{-19} / 6.626 \times 10^{-34} = 6.79 \times 10^{14} \text{ Hz}$ (visible light)

36. Normalization: Show $A = \sqrt{2/L}$ for $\Psi = A \sin(n\pi x/L)$ in 1D box.

$$\int_0^L A^2 \sin^2(n\pi x/L) dx = A^2 \cdot L/2 = 1 \rightarrow A = \sqrt{2/L} \quad \checkmark$$

37. Probability: Probability of finding $n=1$ electron between $x=L/4$ and $x=3L/4$.

$$P = \int_{L/4}^{3L/4} (2/L) \sin^2(\pi x/L) dx = (2/L) \cdot [L/4 + L/2\pi \cdot \sin(\pi/2)] = 0.5 + 1/\pi \approx 0.818 \text{ (81.8\%)}$$

38. Benzene delocalization energy: Using Hückel, $E(\pi) = 2(\alpha+2\beta)+4(\alpha+\beta) = 6\alpha+8\beta$.

Ethylene $E(\pi) = 2(\alpha+\beta)$ each $\rightarrow$ 3 ethylenes $= 6\alpha+6\beta$.

$$\text{Delocalization energy} = (6\alpha+8\beta) - (6\alpha+6\beta) = 2\beta$$

39. Box energy in eV: E_1 for proton ($m=1.673 \times 10^{-27}$ kg) in 0.1 nm box.

$$E_1 = h^2/8mL^2 = (4.39 \times 10^{-67}) / (8 \times 1.673 \times 10^{-27} \times 10^{-20}) = 3.27 \times 10^{-21} \text{ J} = 0.0204 \text{ eV}$$

40. Cyclobutadiene (4π electrons): $4n=4$, $n=1 \rightarrow$ Anti-aromatic. B_2 , B_2 antibonding filled $\rightarrow$ highly unstable. **Hückel anti-aromatic rule confirmed.**

Questions & Answers — 2-Mark and 5-Mark

Exam-oriented questions with model answers

2-Mark Questions

2 MARKS

Q1. Define wave function Ψ . What is Born interpretation?

The wave function $\Psi(x,y,z,t)$ is a mathematical function that completely describes the quantum state of a particle. It contains all information about the particle. According to Born interpretation, the square of the wave function, $|\Psi|^2$, represents the probability density — the probability of finding the particle per unit volume at a given point in space.

2 MARKS

Q2. State Heisenberg's Uncertainty Principle with the mathematical expression.

It is impossible to simultaneously determine both the exact position (Δx) and exact momentum (Δp) of a microscopic particle. The product of their uncertainties satisfies: $\Delta x \cdot \Delta p \geq h/4\pi$. This is a fundamental property of nature, not a measurement limitation.

2 MARKS

Q3. Write the formula for energy of a particle in a 1D box and identify each symbol.

$E_n = n^2 h^2 / (8mL^2)$, where n = quantum number (1,2,3...), h = Planck's constant (6.626×10^{-34} J·s), m = mass of particle (kg), L = length of box (m). Energy is quantized and proportional to n^2 .

2 MARKS

Q4. What is zero-point energy? Why is it non-zero?

Zero-point energy is the energy of the lowest quantum state ($n=1$): $E_1 = h^2/8mL^2$. It is non-zero because of Heisenberg's Uncertainty Principle — if the particle had zero energy, its position and momentum would both be exactly known (zero), which violates HUP. Hence, even at absolute zero, a quantum particle retains minimum energy.

2 MARKS

Q5. Define bond order and give its formula.

Bond order is the effective number of chemical bonds between two atoms in a molecule. Formula: Bond Order = $(N_b - N_a)/2$, where N_b = electrons in bonding MOs, N_a = electrons in antibonding MOs. Higher bond order → stronger, shorter bond. Zero bond order → molecule does not exist.

2 MARKS

Q6. What is LCAO? How many MOs are formed from n AOs?

LCAO (Linear Combination of Atomic Orbitals) is the mathematical method to construct molecular orbitals from atomic orbitals by adding or subtracting wave functions: $\Psi_{\text{Bonding}} = \Psi_A + \Psi_B$ and $\Psi_{\text{Antibonding}} = \Psi_A - \Psi_B$. From n atomic orbitals, exactly n molecular orbitals are formed (conservation of orbitals).

2 MARKS

Q7. Why is O₂ paramagnetic? Explain using MOT.

O₂ has 16 electrons. In the MO filling, the last 2 electrons go into the degenerate π^*2p orbitals — by Hund's rule, one electron occupies each π^*2p orbital with parallel spins ($\uparrow\uparrow$). These 2 unpaired electrons make O₂ paramagnetic (attracted by magnetic field). VBT incorrectly predicts diamagnetic — MOT correctly explains this.

2 MARKS

Q8. State Hückel's rule for aromaticity.

A planar, cyclic, fully conjugated molecule is aromatic if it contains $(4n+2)$ π electrons, where $n = 0, 1, 2, 3, \dots$. Examples: benzene ($n=1$, 6π electrons), naphthalene ($n=2$, 10π electrons). Anti-aromatic: $4n$ π electrons (unstable). Aromatic molecules have exceptional stability due to delocalization energy.

2 MARKS

Q9. Predict whether He₂ can exist. Justify.

He₂ cannot exist. It has 4 electrons: config $(\sigma 1s)^2(\sigma^* 1s)^2$. $N_b=2$, $N_a=2$. Bond Order = $(2-2)/2 = 0$. Zero bond order means no net bonding. The bonding effect of 2 electrons in $\sigma 1s$ is exactly cancelled by the 2 antibonding electrons in $\sigma^* 1s$. Hence He₂ is unstable and does not exist.

2 MARKS

Q10. How does bond length vary with bond order?

Bond length decreases as bond order increases. More bonding electrons mean the atoms are held closer together. Example: N₂ (BO=3, bond length=110 pm) < O₂ (BO=2, bond length=121 pm) < F₂ (BO=1, bond length=142 pm). Higher bond order also means greater bond strength (dissociation energy).

5-Mark Questions

5 MARKS

Q11. Derive the expression for energy levels of a particle in a 1D box with detailed steps.

Setup: Particle confined in 1D box of length L . $V=0$ inside ($0 < x < L$), $V=\infty$ at walls.

Step 1 – Schrödinger Equation (inside):

$$-(\hbar^2/2m)(d^2\Psi/dx^2) = E\Psi \rightarrow d^2\Psi/dx^2 = -k^2\Psi, \text{ where } k^2 = 2mE/\hbar^2$$

Step 2 – General Solution:

$$\Psi(x) = A \sin(kx) + B \cos(kx)$$

Step 3 – Boundary Condition at $x=0$:

$$\Psi(0) = 0 \rightarrow B \cos(0) = 0 \rightarrow B = 0 \rightarrow \Psi = A \sin(kx)$$

Step 4 – Boundary Condition at $x=L$:

$$\Psi(L) = 0 \rightarrow A \sin(kL) = 0 \rightarrow kL = n\pi \rightarrow k = n\pi/L, n=1,2,3,\dots$$

Step 5 – Energy:

$$k^2 = 2mE_n/\hbar^2 \rightarrow (n\pi/L)^2 = 2mE_n/\hbar^2 \rightarrow E_n = n^2\pi^2\hbar^2/(2mL^2) = n^2h^2/(8mL^2)$$

Step 6 – Wave function:

$$\Psi_n(x) = \sqrt{2/L} \cdot \sin(n\pi x/L) \text{ (after normalization)}$$

5 MARKS

Q12. Explain Molecular Orbital Theory with LCAO method. What are bonding and antibonding MOs?

MOT Basics: Electrons in a molecule occupy molecular orbitals (MOs) extending over the entire molecule, formed by combining atomic orbitals.

$$\text{LCAO: } \Psi(\text{MO}) = c_1\Psi_A \pm c_2\Psi_B$$

Addition (constructive): $\Psi_{\text{Bonding}} = \Psi_A + \Psi_B \rightarrow$ increased electron density between nuclei $\rightarrow$ lower energy (stabilizing)

Subtraction (destructive): $\Psi_{\text{Antibonding}} = \Psi_A - \Psi_B \rightarrow$ nodal plane between nuclei $\rightarrow$ higher energy (destabilizing)

Conditions for LCAO: (1) Similar energy, (2) Good overlap, (3) Compatible symmetry

Bonding MO: Energy lower than constituent AOs; electron density concentrated between nuclei; sigma (head-on) or pi (side-on) overlap

Antibonding MO: Energy higher than AOs; nodal plane between nuclei; denoted σ^* , π^* ; destabilizes molecule

Bond Order = $(N_b - N_a)/2$. Conservation: $n \text{ AOs} \rightarrow n \text{ MOs}$

5 MARKS

Q13. Draw and explain the MO energy level diagram of O_2 . Show why it is paramagnetic.

O_2 has 16 electrons. MO energy order for O_2 ($Z \geq 8$):

$\sigma_{1s} < \sigma^*_{1s} < \sigma_{2s} < \sigma^*_{2s} < \sigma_{2p} < \pi_{2p} = \pi_{2p} < \pi^*_{2p} = \pi^*_{2p} < \sigma^*_{2p}$

Configuration: $(\sigma_{1s})^2(\sigma^*_{1s})^2(\sigma_{2s})^2(\sigma^*_{2s})^2(\sigma_{2p})^2(\pi_{2p})^2(\pi_{2p})^2(\pi^*_{2p})^1(\pi^*_{2p})^1$

Bond Order: $N_b=10, N_a=6 \rightarrow B.O.=(10-6)/2=2$ (double bond ✓)

Paramagnetism: The 16th and 15th electrons enter the degenerate π^*_{2p} pair. By Hund's rule, one electron goes to each orbital with parallel spins. These 2 unpaired electrons create a permanent magnetic dipole $\rightarrow O_2$ is paramagnetic.

This is a major triumph of MOT over VBT (which incorrectly predicts diamagnetic O_2).

5 MARKS

Q14. Discuss π -molecular orbitals in benzene and explain its aromatic stability.

Benzene (C_6H_6): 6 carbons in a ring, each with one p orbital perpendicular to molecular plane $\rightarrow$ 6 π -MOs formed.

Energy levels (Hückel):

ψ_1 : $E = \alpha + 2\beta$ (lowest, 0 nodes, all-bonding)

ψ_2, ψ_3 : $E = \alpha + \beta$ (degenerate pair, 1 node each)

ψ_4, ψ_5 : $E = \alpha - \beta$ (degenerate pair, antibonding)

ψ_6 : $E = \alpha - 2\beta$ (highest, 3 nodes)

Electron filling: 6 π electrons fill ψ_1, ψ_2, ψ_3 (all bonding MOs). All antibonding MOs empty.

Delocalization energy: 2β more stable than 3 isolated double bonds $\rightarrow$ extra stability (aromatic stabilization).

Hückel's rule: $6 = 4(1) + 2$ ($n=1$) $\rightarrow$ aromatic ✓. This explains benzene's unusual stability, equal bond lengths (139 pm), and preference for substitution over addition reactions.

Quick Revision Sheet

All key formulas, concepts, and tricks in one place

Key Formulas at a Glance

de Broglie Wavelength

$$\lambda = h/(mv) = h/p$$

Smaller mass or lower velocity $\rightarrow$ larger wavelength

Heisenberg Uncertainty

$$\Delta x \cdot \Delta p \geq h/4\pi$$

$$\text{Also: } \Delta E \cdot \Delta t \geq h/4\pi$$

Particle in Box Energy

$$E_n = n^2 h^2 / (8mL^2)$$

$$E_2 = 4E_1, E_3 = 9E_1, E_4 = 16E_1$$

$$\text{Nodes in } \Psi_n = n-1$$

Bond Order

$$\text{B.O.} = (N_b - N_a)/2$$

$$\uparrow \text{ B.O.} = \uparrow \text{ Bond strength, } \downarrow \text{ Bond length}$$

$$\text{B.O.} = 0 \rightarrow \text{molecule doesn't exist}$$

LCAO Combinations

$$\Psi_B = \Psi_A + \Psi_B \text{ (bonding, lower E)}$$

$$\Psi_{AB} = \Psi_A - \Psi_B \text{ (antibonding, higher E, has node)}$$

$$n \text{ AOs} \rightarrow n \text{ MOs always}$$

Planck's Equation

$$E = h\nu = hc/\lambda$$

$$h = 6.626 \times 10^{-34} \text{ J}\cdot\text{s}$$

$$c = 3 \times 10^8 \text{ m/s}$$

Memory Tricks & Mnemonics

MO Order for O_2, F_2 ($Z \geq 8$)

MO Order for N_2 ($Z \leq 7$)

Aromaticity Rule

"Sassy Sigmas Put Pairs Perfectly"

$\sigma 1s, \sigma^* 1s, \sigma 2s, \sigma^* 2s, \sigma 2p, \pi 2p \times 2, \pi^* 2p \times 2, \sigma^* 2p$

Same but $\pi 2p$ comes BEFORE $\sigma 2p$

$\dots \sigma^* 2s, \pi 2p = \pi 2p, \sigma 2p, \pi^* 2p = \pi^* 2p, \sigma^* 2p$

" $4n+2$ = Aromatic (Happy!)"

" $4n$ = Anti-aromatic (Sad!)"

Benzene: $n=1 \rightarrow 6\pi \checkmark$

Important Constants

Constant	Symbol	Value	Unit
Planck's constant	h	6.626×10^{-34}	J·s
Reduced Planck's	$\hbar = h/2\pi$	1.055×10^{-34}	J·s
Speed of light	c	3×10^8	m/s
Electron mass	m_e	9.109×10^{-31}	kg
Proton mass	m_p	1.673×10^{-27}	kg
1 eV in Joules	1 eV	1.602×10^{-19}	J

Quick Bond Order Reference

Molecule	e ⁻	B.O.	Magnetic	Species	e ⁻	B.O.	Magnetic
H ₂	2	1	Dia	H ₂ ⁺	1	0.5	Para
N ₂	14	3	Dia	He ₂	4	0	—
O ₂	16	2	Para	O ₂ ⁺	15	2.5	Para
F ₂	18	1	Dia	O ₂ ⁻	17	1.5	Para
CO	14	3	Dia	O ₂ ²⁻	18	1	Dia
NO	15	2.5	Para	NO ⁺	14	3	Dia

Comparison Tables & Summary

MOT vs VBT, Homonuclear vs Heteronuclear, and conceptual summaries

18.1 MOT vs Valence Bond Theory (VBT)

Aspect	Valence Bond Theory (VBT)	Molecular Orbital Theory (MOT)
Basic concept	Electrons localized in atomic orbitals; bond from overlap of AOs of two atoms	Electrons delocalized in molecular orbitals extending over entire molecule
Electron arrangement	Electrons stay near their original atoms	Electrons belong to all atoms equally
Type of orbitals	Atomic orbitals (s, p, d, f) and hybrid orbitals	Molecular orbitals (σ , π and their antibonding counterparts)
Delocalization	Explained by resonance structures (artificial)	Naturally explained by delocalized MOs
O ₂ magnetism	Fails — predicts diamagnetic (wrong!)	Correct — predicts paramagnetic ✓
Bond order	Whole numbers only (1, 2, 3)	Can be fractional (0.5, 1.5, 2.5)
Prediction power	Limited; fails for many molecules	Powerful; works for all molecules
Benzene bonding	Resonance structures needed	Directly explained by π -MO delocalization
Complexity	Simpler, intuitive	More complex mathematically
Antibonding orbitals	Not explicitly considered	Explicitly includes both bonding and antibonding

18.2 Homonuclear vs Heteronuclear Diatomic Molecules

Aspect	Homonuclear (A ₂)	Heteronuclear (AB)
Examples	H ₂ , N ₂ , O ₂ , F ₂ , Cl ₂	CO, NO, HF, HCl
Atomic orbital energies	Identical (same atoms)	Different (different electronegativities)

MO character	Equal contribution from both atoms	Unequal contribution; more electronegative atom dominates lower-energy MO
Bond polarity	Non-polar (symmetric)	Polar bond (dipole moment)
LCAO coefficients	$c_1 = c_2$ (equal mixing)	$c_1 \neq c_2$ (unequal mixing)
HOMO/LUMO location	Delocalized equally over both atoms	HOMO more on one atom (e.g., CO: HOMO on C)
Example — CO	—	CO: isoelectronic with N_2 ($14e^-$), $BO=3$, but HOMO concentrated on C → CO is Lewis base through C

18.3 Classical vs Quantum Mechanics — Key Differences

Feature	Classical Mechanics	Quantum Mechanics
Applicable to	Macroscopic objects (g, kg range)	Microscopic particles (atoms, electrons)
Energy	Continuous (any value allowed)	Quantized (only specific values)
Trajectories	Definite path (position + velocity known)	No definite path; only probability distribution
Wave-particle duality	Light is wave OR particle	All matter has both wave AND particle nature
Measurement	Exact values of all properties simultaneously	HUP: cannot know position AND momentum exactly
Key equations	Newton's $F = ma$	Schrödinger's $\hat{H}\Psi = E\Psi$
Electron location	Definite orbit (Bohr model)	Probability cloud (orbital = probability region)
Zero-point energy	Zero (particle can be at rest)	Non-zero (always has minimum energy)

18.4 Concept Summary — Unit I at a Glance



The Story of Unit I

- Classical physics failed to explain atomic stability, blackbody radiation, photoelectric effect → Needed quantum theory
- Planck (1900): Energy is quantized ($E=h\nu$); Einstein (1905): Light has particle nature (photons)
- de Broglie (1924): Matter has wave nature ($\lambda=h/mv$); Heisenberg (1927): Uncertainty Principle
- Schrödinger (1926): Wave equation $\hat{H}\Psi=E\Psi$ describes quantum particles; $|\Psi|^2$ = probability density
- Particle in 1D box: $E_n=n^2h^2/8mL^2$ — simplest example of energy quantization with zero-point energy

- MOT + LCAO: AOs combine to form bonding and antibonding MOs; fill by Aufbau, Pauli, Hund's rules
- Bond order = $(N_b - N_a)/2$ determines stability, bond strength, and magnetic character
- π -MOs in conjugated systems (butadiene: 4 MOs; benzene: 6 MOs) explain delocalization and aromaticity ($4n+2$ rule)



Final Exam Tips

(1) Always write MO configuration in order before calculating bond order. (2) Remember the MO energy order swap between N_2 and O_2 . (3) For particle-in-box problems, use $E_n = n^2 E_1$ to save time. (4) Check paramagnetism by counting unpaired electrons after MO filling. (5) For aromaticity: check cyclic + conjugated + $4n+2$ π electrons. All three conditions must be satisfied.

End of Unit I — Structure and Bonding Models

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